

POWERFACTORY

PowerFactory 2024

Technical Reference

Static Synchronous Series Compensation

ElmSssc

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

A Static Synchronous Series Compensator (SSSC) is a device capable of injecting a series compensating voltage in series with a transmission line in order to control the power flow or the current. It can provide both inductive and capacitive compensation, depending on the phase relationship between the injected voltage and the line current. Differently from other impedance based series compensation schemes, the magnitude of the injected compensating voltage of an SSSC is independent of the line-current. Various configurations of SSSC have been proposed. *PowerFactory* offers two models:

- Voltage source
- Two-level converter

The *Voltage source* model is shown in Figure 1.1. It consists of an ideal voltage source in series with an impedance in each phase. The voltage sources can be independently controlled on each phase. The injected voltage V is always in quadrature with the line current. It can be used to represent different configurations for an SSSC.

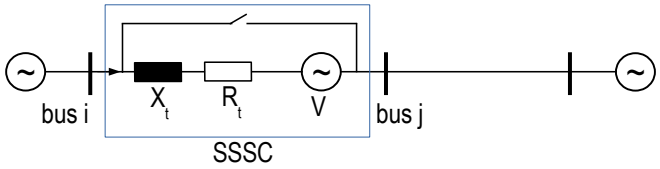


Figure 1.1: Single-line diagram of the voltage source SSSC model

The *Two-level converter* model is shown in Figure 1.2. A PWM two-level converter injects the compensating voltage in series with the line through a booster transformer, whose type (*TypTrb*) can be chosen on the *Basic Data* page. The injected voltage is not exactly in quadrature with the line current, in order to balance the losses in the PWM converter and its series reactor. Note that the booster transformer representation is simplified and it does not include the positive magnetizing branch.

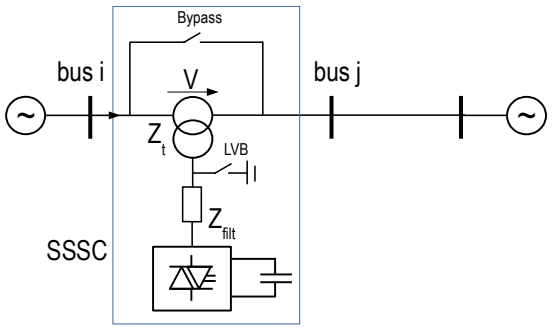


Figure 1.2: Single-line diagram of the two-level converter SSSC model

For the two-level converter model, the angle between V and the line current is close to but not exactly $90°$, due to the fact that the internal converter losses must be supplied from the grid.

The *PowerFactory* SSSC model uses the convention that a positive voltage (or reactance) represent an inductive injection, while a negative voltage (or reactance) represent a capacitive injection.

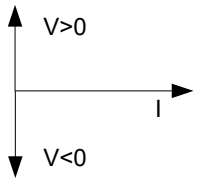


Figure 1.3: Meaning of positive/negative voltage

Both models can be bypassed, by selecting the *Bypassed* flag in the *Basic Data* page.

2 Voltage Source Model

The Voltage source model can be operated:

- to inject only positive sequence voltage
- to inject an independent voltage on each phase (only for unbalanced calculations)

In order to inject an independent voltage on each phase during load flow calculations, the flag *i_phwise* must be selected.

When the Voltage source model is operated to inject only a positive sequence voltage, the positive, negative and zero sequence equivalent circuit of the voltage source SSSC model is shown in Figures 2.1 and 2.2.

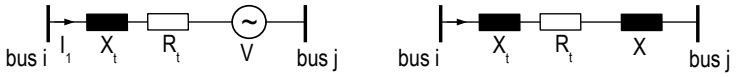


Figure 2.1: Positive sequence equivalent circuit of the voltage source SSSC model, no phase-wise control. Voltage injection representation (left) and reactance representation (right)

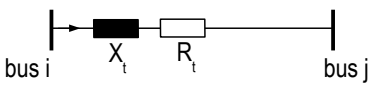


Figure 2.2: Negative and zero sequence equivalent circuit of the voltage source SSSC model, no phase-wise control

When phase-wise control is selected the SSSC is better represented (only for unbalanced calculations) by an independent injection on each phase, as shown in Figure 2.3.

Independent voltage (and reactance) input per phase are also available to use in *(*QDSL) and *(*DSL) models.

In balanced calculations, the Voltage source SSSC model is always represented as in Figure 2.1.

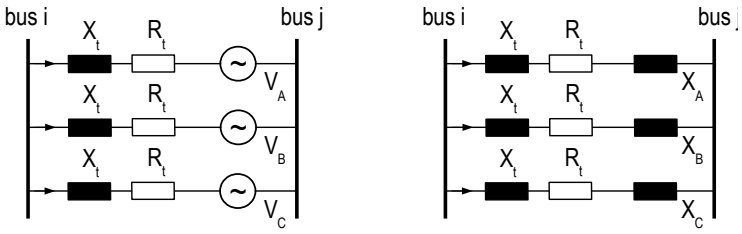


Figure 2.3: Equivalent circuit of the voltage source SSSC model, phase-wise control. Voltage injection representation (left) and reactance representation (right)

3 Two-Level Converter Model

In fundamental frequency calculations, the two-level converter model can only inject a positive sequence voltage. In EMT simulations it can also inject a negative sequence voltage. It is not possible to inject an independent voltage on each phase, hence the option for phase-wise control (*i_phwise*) is not available.

For the two-level converter model, the angle between V and I_1 is close to but not exactly $90°$, due to the fact that the internal converter losses must be supplied from the grid.

The positive, negative and zero sequence equivalent circuit of the two-level converter SSSC model are shown in Figures 3.1, 3.2 and 3.3.

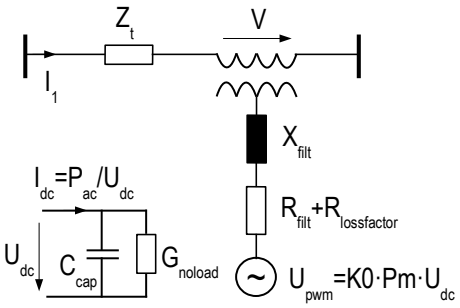


Figure 3.1: Positive sequence equivalent circuit of the two-level converter SSSC model

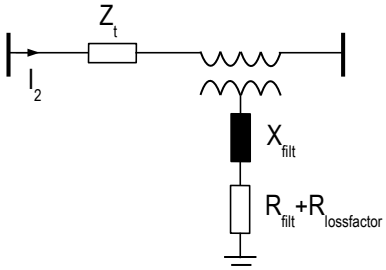


Figure 3.2: Negative sequence equivalent circuit of the two-level converter SSSC model

The representation of the booster transformer has been simplified and the positive sequence magnetization is not considered. The positive sequence leakage impedance is modelled at the HV side, without considering the leakage distribution factor. The zero sequence magnetization and leakage distribution factor are instead considered.

The zero sequence voltage cannot flow on the LV side (i.e. the PWM side) of the booster

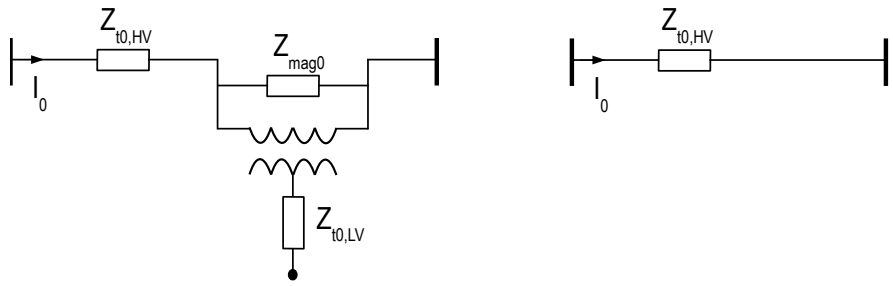


Figure 3.3: Zero sequence equivalent circuit of the two-level converter SSSC model. Booster transformer LV side is Y (left) or D (right)

transformer, since the PWM converter is assumed to be ungrounded.

Active power conservation between AC- and DC-side is assumed, leading to:

$$P_{AC} = \Re(\sqrt{3} \cdot \underline{U}_{AC} \cdot \underline{I}^*_{AC}) = U_{DC} \cdot I_{DC} = P_{DC} \quad (1)$$

$$P_{DC} = U_{DC} \cdot I_{cap} + G_{noload} \cdot U^2_{DC} \quad (2)$$

where:

- P_{AC}: PWM converter AC-active power
- P_{DC}: PWM converter DC-power
- $\underline{U}_{AC}$: PWM converter AC-voltage phasor (RMS-value)
- U_{DC}: PWM converter DC-voltage
- $\underline{I}^*_{AC}$: PWM converter conjugate complex value of AC-current phasor (RMS-value)
- I_{DC}: PWM converter DC-current
- I_{cap}: PWM converter DC capacitor current

Converter losses are represented by the no-load losses and a resistive loss factor (G_{noload} and $R_{lossfactor}$ in Figure 3.1). Note that differently from the PWM converter model (*ElmVsc*), the losses due to the resistive loss factor are represented on the AC side. Also, switching losses are not considered in the SSSC model. The no-load losses conductance is calculated as:

$$G_{noload} = \frac{Pnold}{1000 \cdot unomdc^2}$$

where $Pnold$ and $unomdc$ are defined in the *Basic Data - Converter page*.

3.1 AC Side Representation

At fundamental frequency, the PWM converter is modelled by a DC-voltage controlled AC-voltage source. For values of $|Pm| < 1$ (where $|Pm|$ is the modulation index magnitude), the following equations can be applied:

$$U_{ACr} = K_0 \cdot Pmr \cdot U_{DC}$$
$$U_{ACi} = K_0 \cdot Pmi \cdot U_{DC} \quad (3)$$

The pulse width modulation index, which can be defined by real- and imaginary part or by d- and q- axis components is the (complex) control variable of the PWM-converter. The converter supports only sinusoidal modulation and the K_0 factor is defined as:

$$K_0 = \frac{\sqrt{3}}{2 \cdot \sqrt{2}} \quad (4)$$

In all steady-state functions and RMS-simulation, real and imaginary parts of the AC-voltage correspond to the positive sequence component. In EMT-simulations, they correspond to the space-phasor.

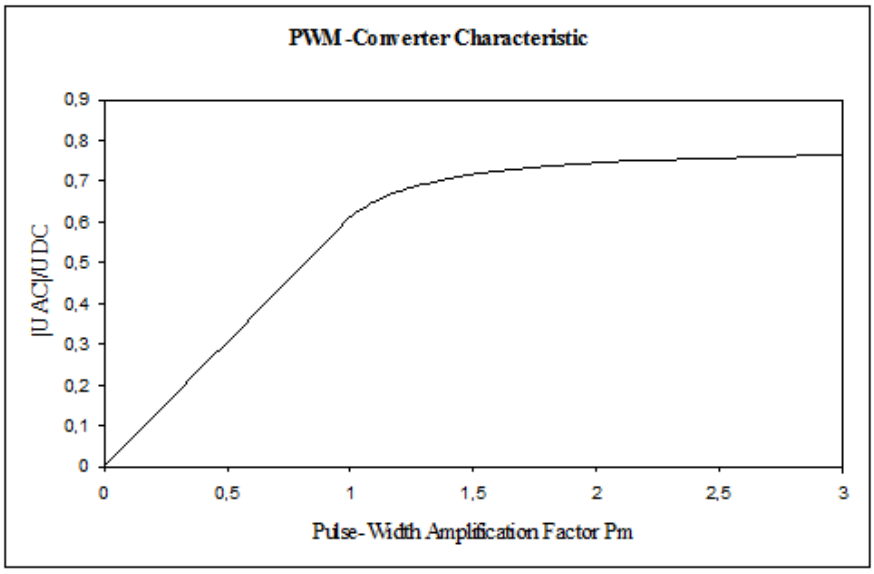


Figure 3.4: PWM-converter saturation

In case of $|Pm| > 1$ and with sinusoidal PWM modulation, the converter starts saturating resulting in the fundamental frequency relationship between the AC-RMS-value and the DC-voltage according to Figure 3.4. As an additional result of converter saturation, low order harmonics start to increase and thus only low levels of saturation are usually allowed.

To extend the linear range of operation of the converter and for a better utilization of the DC-link voltage, the sinusoidal modulation technique can be modified by the injection of a third harmonic in the modulating signals. The third harmonic will not be present in the grid line-line voltages. Four options are available for the third harmonic injection:

- None.
- Sinusoidal, 1/6 amplitude.
- Sinusoidal, 1/4 amplitude.
- Equivalent to SVPWM.

Choosing ”None”, sinusoidal modulation will be implemented. Sinusoidal modulation with third harmonic injection of type ”Sinusoidal, 1/6 Amplitude” is often referred in the literature as THIPWM. The fourth option results in a modulation equivalent to the space vector modulation technique.

The original sinusoidal modulating signal are given as:

$$m_{a,sin} = A \cdot cos(\omega \cdot t)$$
$$m_{b,sin} = A \cdot cos(\omega \cdot t - 2/3\pi)$$
$$m_{c,sin} = A \cdot cos(\omega \cdot t + 2/3\pi)$$

The resulting modulating signals with third harmonic injection are given as:

$$m_a = m_{a,sin} + v_0$$
$$m_b = m_{b,sin} + v_0$$
$$m_c = m_{c,sin} + v_0$$

where v_0 depends on the selected third harmonic injection method.

For ”Sinusoidal, 1/6 Amplitude”:

$$v_0 = A/6 \cdot cos(3 \cdot \omega \cdot t)$$

For ”Sinusoidal, 1/4 Amplitude”:

$$v_0 = A/4 \cdot cos(3 \cdot \omega \cdot t)$$

For ”Equivalent to SVPWM”:

$$v_0 = \frac{max(m_{a,sin},m_{b,sin},m_{c,sin}) + min(m_{a,sin},m_{b,sin},m_{c,sin})}{2}$$

A modulating signal with different types of third harmonic injection is shown in figure 3.5.

The fundamental frequency relationship between the AC-RMS-value and the DC-voltage according to Figure 3.4 is no longer valid if third harmonic injection is used. Depending on the type of third harmonic injection selected, in the overmodulation region the fundamental frequency relationship between AC-RMS-value and the DC-voltage is modified according to figure 3.6.

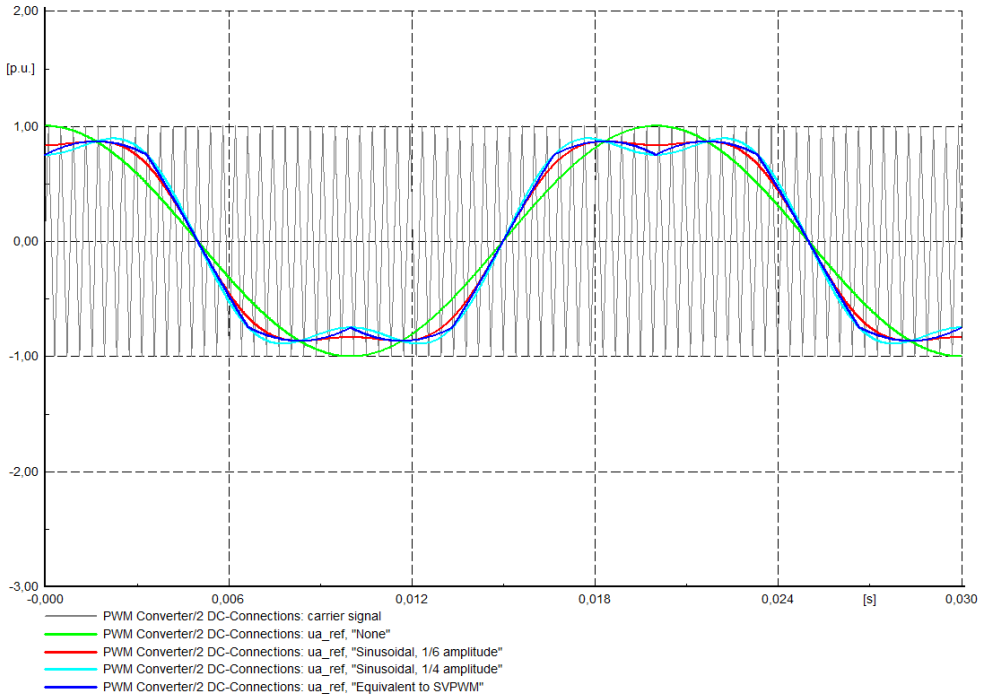


Figure 3.5: PWM Converter reference signal. Pulse width modulation index is equal to 1

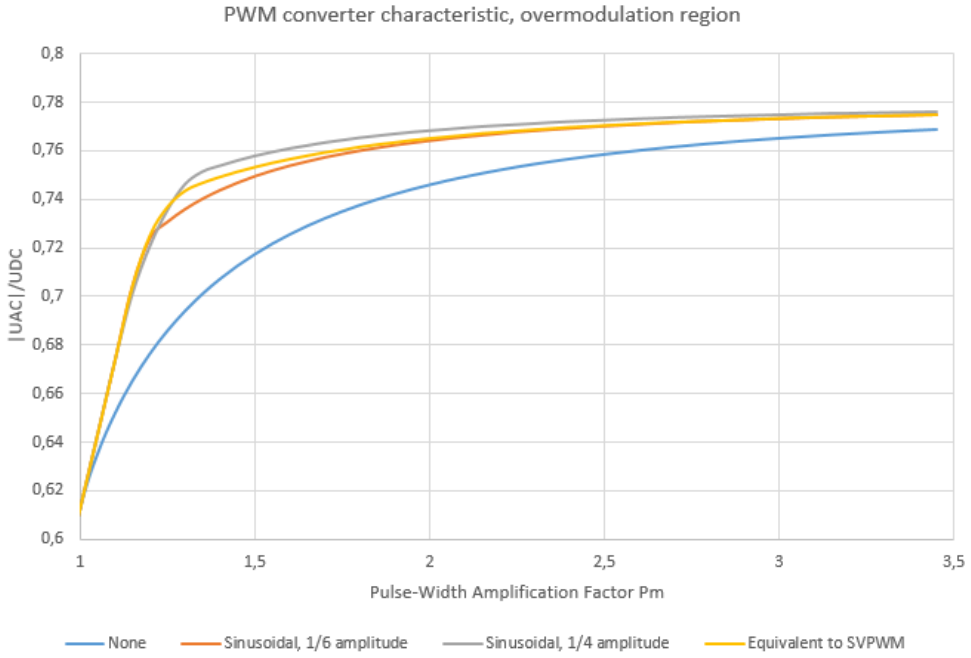


Figure 3.6: PWM-converter saturation with third harmonic injection

4 Load Flow

The SSSC model supports six different control modes, each of them is described in detail in the next sections:

- Active power
- Current magnitude
- Effective reactance
- Voltage injection
- Current droop
- Monitoring

4.1 Voltage and Current Limits

Minimum and maximum voltage and operating current limits can be defined in the *Load Flow* page. The SSSC operates in the selected control modes if: a) the injected voltage, V in Figures 2.1 and 3.1, is within the defined minimum and maximum voltage limits and b) the SSSC current, I_1 in Figures 2.1 and 3.1, is within minimum and maximum operating current limits.

When the injected voltage V reaches the minimum or maximum voltage limit, the control mode is switched to constant voltage. The previous control mode is resumed if the system conditions change and allow operation within the voltage control limits.

When the SSSC current I_1 reaches the minimum or maximum operating current limit, the SSSC is bypassed, i.e. the bypass switch in Figures 1.1 and 1.2 is closed. The previous control mode can no longer be resumed.

For the Voltage source model, it is possible to perform the control phase-wise, by selecting the *i_phwise* option. In this case, the voltage and current limit are checked for each phase separately, using the phase voltages and currents instead of the positive sequence values. In case one phase current is outside of the operating current limits, the SSSC is bypassed on all phases.

4.2 Active Power Control

When *Active power* control is selected, the SSSC controls the positive sequence power or, in case of phase-wise control, the power of each phase to the defined setpoint value.

In case of phase-wise control the user defines the power setpoint $psetp$ for one phase. Therefore, when running a balanced load flow the positive sequence power will be controlled to $3 \cdot psetp$.

The active power can be controlled either at terminal i or j, see Figure 1.1.

Alternatively the power can be controlled at a remote cubicle. In this case, the *Orientation* must be defined. The information on the orientation is used to resume *Active power* control, once the SSSC entered the voltage limit mode. A positive orientation *+dP/dXcap* means that an increase of the equivalent capacitive reactance of the SSSC causes an increase of power flow (absolute value) at the controlled cubicle. Figure 4.1 shows how to define the sign of *P setpoint* and the orientation. For example, if power is controlled at cubicle 1, an increase of the capacitive reactance of the SSSC will cause less power to flow through cubicle 1 (and more power flow through the SSSC itself). Same is valid for cubicle 2.

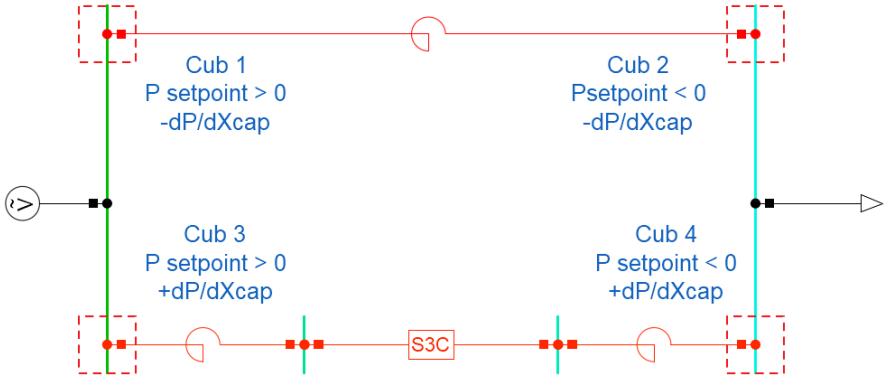


Figure 4.1: Definition of active power setpoint and orientation for remote control

4.3 Current Magnitude Control

When *Current magnitude* control is selected, the SSSC controls the positive sequence current or, in case of phase-wise control, the current of each phase to the defined setpoint value.

4.4 Effective Reactance Control

When *Effective reactance* control is selected, the SSSC controls the equivalent injected positive sequence reactance or, in case of phase-wise control, the injected reactance in each phase to the defined setpoint value.

A positive value indicates an inductive reactance, while a negative value indicates a capacitive reactance.

For phase-wise control, a different reactance setpoint per phase can be defined. In this case, when executing a balanced load flow the reactance will be controlled to the average value of the three defined reactance setpoints.

With reference to Figure 2.1, the equivalent positive sequence reactance is defined as:

$$X = Im\left(\frac{\underline{V}}{\underline{I_1}}\right)$$

and the per-phase reactance is defined as:

$$X_A = Im\left(\frac{\underline{V_A}}{\underline{I_A}}\right)$$
$$X_B = Im\left(\frac{\underline{V_B}}{\underline{I_B}}\right)$$
$$X_C = Im\left(\frac{\underline{V_C}}{\underline{I_C}}\right)$$

For the Voltage source model, the injected voltage V is always exactly $90°$ lagging/leading the

current I_1. For the two-level converter model, the angle between V and I_1 is close to but not exactly $90°$, due to the fact that the internal converter losses must be supplied from the grid.

4.5 Voltage Injection Control

When *Voltage injection* control is selected, the SSSC controls the injected positive sequence voltage or, in case of phase-wise control, the injected voltage in each phase to the defined setpoint value.

A positive value indicates the voltage leads the line current (equivalent inductive reactance), while a negative value indicates the voltage lags the line current (equivalent capacitive reactance).

For phase-wise control, a different voltage setpoint per phase can be defined. In this case, when executing a balanced load flow the injected voltage will be controlled to the average value of the three defined voltage setpoints.

For the Voltage source model, the injected voltage V is always exactly $90°$ lagging/leading the current I_1. For the two-level converter model, the angle between V and I_1 is close to but not exactly $90°$, due to the fact that the internal converter losses must be supplied from the grid.

4.6 Current Droop Control

When *Current droop* control is selected, the SSSC controls the injected positive sequence voltage or, in case of phase-wise control, the injected voltage in each phase according to the following equations:

$$V = \begin{cases} kdroop_ind \cdot (I - isetp_ind) & I > isetp_ind \\ 0 & isetp_cap < I < isetp_ind \\ kdroop_cap \cdot (I - isetp_cap) & I < isetp_cap \end{cases} \quad (7)$$

where

- $isetp_ind$ is the inductive current setpoint in kA, *Iset (ind.)*
- $isetp_cap$ is the capacitive current setpoint in kA, *Iset (cap.)*
- $kdroop_ind$ is the inductive droop in V/A, *Droop (ind.)*
- $kdroop_cap$ is the capacitive droop in V/A, *Droop (cap.)*

Once the current droop is engaged in inductive mode, the SSSC will inject the voltage according to Equation 7 until the current $I >= isetp_ind - ioffset_ind$. If the current decreases further, the injected voltage is zero.

Once the current droop is engaged in capacitive mode, the SSSC will inject the voltage according to Equation 7 until the current $I <= isetp_cap + ioffset_cap$. If the current increases further, the injected voltage is zero.

For the two-level converter model, Equation 7 defines the inductive/capacitive part of the injected voltage. A resistive part will also be injected in order to supply the internal losses of the converter model.

Note that when the injected voltage is zero, the series impedance $R_t + jX_t$ (or the booster transformer in case of a two-level converter model) is still inserted in series with the network, i.e. the SSSC is not bypassed.

4.7 Monitoring Control

When in *Monitoring* control, the SSSC does not inject any voltage. The series impedance (or the booster transformer impedance) is still connected in series with the grid.

4.8 Switch to Current Droop Control

If the selected control mode is *Active power*, *Effective reactance*, *Voltage injection* or *Monitoring*, it is possible to select the *Switch to current droop control* option, parameter *i_override*. The SSSC will remain in the original control mode if $isetp_cap < I < isetp_ind$, otherwise it will switch to a *Current droop* control mode, injecting a voltage according to 7. Once the current droop is engaged in inductive mode, the SSSC will inject the voltage according to Equation 7 until the current $I >= isetp_ind - ioffset_ind$. Once the current droop is engaged in capacitive mode, the SSSC will inject the voltage according to Equation 7 until the current $I <= isetp_cap + ioffset_cap$. If the current returns within the limits $isetp_cap + ioffset_cap < I < isetp_ind - ioffset_ind$, the SSSC will switch back to its original control mode.

The calculation parameter *Vpre_override* stores the value of the injected voltage before switching to current droop control (for unbalanced calculations, *Vpre_override_A*, *Vpre_override_B*, *Vpre_override_C*).

It may happen that when the condition to switch to current droop control are fulfilled, e.g. $I >= isetp_ind$, the voltage injected according to Equation 7 is less than the voltage injection in the original control mode (i.e. *Vpre_override*) due to a low value of the parameter *kdroop_ind*. In this case, switching to current droop control would cause an even further increase of the SSSC current. In this case the SSSC will continue to inject the voltage calculated according to the original control mode.

4.9 QDSL input

The SSSC model can be controlled in load flow via a QDSL model. Available input for balance load flow are:

- *Xin*
- *Vin*
- *pset*

and for unbalanced load flow:

- *Xin_A*, *Xin_B*, *Xin_C*
- *Vin_A*, *Vin_B*, *Vin_C*
- *pset*

In order for the input to be considered, the corresponding control model must be selected in the element (e.g. when using the input *Xin* the parameter *i_ctrl* must be set to ”Effective reactance”). Otherwise a warning message is printed in the output window.

Notice that if the flags *Switch to current droop control* or *Operating current limits* are selected in the element, they will still be considered when providing a QDSL input.

4.10 Additional Calculation Parameter

When the *Switch to current droop control* is selected, the control mode of the SSSC may change during a load flow calculation. Moreover, the SSSC can pass to a bypassed state if minimum or maximum operating current limits are reached. The parameter *settlingMode* (for unbalanced calculation *settlingMode_A*, *settlingMode_B*, *settlingMode_C*) informs the user on the final control mode of operation of the SSSC according to the following table:

Value	Meaning
0	Bypassed, Monitoring with *i_override = 0*
1	Voltage control
2	Reactance control
3	Current droop control, Monitoring with *i_override = 1* (not engaged)
4	Current droop control, inductive (engaged)
5	Current droop control, capacitive (engaged)
6	Bypassed, max. operating current limit reached
7	Bypassed, min. operating current limit reached
8	Active power control
9	Current magnitude control

Table 4.1: Definition of calculation parameter *settlingMode*

4.11 Two-Level Converter Model

The model supports both direct and indirect control of the two-level converter. In case of *direct control*, the user specifies a DC voltage setpoint and the PWM factor needed to obtain the control objective is calculated during the load flow. This is the typical way a PWM converter connected to the grid is controlled. In case of *indirect control*, the user specifies the setpoint for the PWM factor and the AC voltage is controlled by varying the DC voltage (the AC voltage is proportional to the DC voltage). The required DC voltage is calculated during the load flow. In dynamic simulations, the variation of the DC voltage is achieved by varying transiently the angle of the SSSC injected voltage compared to the line current, which causes a transient active power exchanged with the grid.

4.11.1 Consider Modulation Index Limit

The option is only available for *direct control*.

For a given control mode, the solution of the Load Flow calculation will satisfy the control objective for the PWM converter, disregarding the physical limits due to the actual DC voltage. To account for these limits, the flag *Consider modulation index limit* should be checked and a maximum PWM factor should be specified. In this case, the load flow solution will limit the pulse modulation index to the specified value and the control objective might not be achieved. A warning message is printed in the output window.

5 RMS and EMT simulation

The *bypass* input is available both for the *Voltage source* model and for the *Two-level converter* model. Setting its value to 1 is equivalent to closing the bypass switch in Figure 1.1 or Figure 1.2.

Output *V*, injected voltage, and *X*, effective reactance, are available also for both models.

In EMT simulations only it is possible to define an external submodel in which a detailed representation of the SSSC can be created. The submodel replaces the internal model. A template ElmSssc submodel containing the necessary submodel connectors can be created by selecting the target ElmSssc, right clicking the mouse and selecting *Network Models*, *Submodels*, *New...*.

5.1 Voltage Source Model

The *Voltage source* model can be controlled with the following set of input for balanced RMS simulation:

- *Xin*
- *Vin*
- *u1r_in*, *u1i_in*

and for unbalanced RMS simulation:

- *Xin_A*, *Xin_B*, *Xin_C*
- *Vin_A*, *Vin_B*, *Vin_C*
- *u1r_in*, *u1i_in*, *u2r_in*, *u2i_in*, *u0r_in*, *u0i_in*

When *Xin* or *Vin* are used, a positive value means the SSSC is injecting an inductive voltage (voltage leading the current by $90°$) and a negative value means the SSSC is injecting a capacitive voltage (voltage lagging the current by $90°$). When using *u1r_in, u1i_in* the user can vary the angle of the injected voltage compared to the current.

For EMT simulation the *Voltage source* model can be controlled with the following input:

- *Vin_A*, *Vin_B*, *Vin_C*

Note that in EMT simulations, *Vin_A*, *Vin_B* and *Vin_C* must be defined via an external control model as sinusoidal functions of time. The user has to take care in the external control model to properly align the voltages with respect to the line current.

5.2 Two-Level Converter Model

The dynamics of the DC capacitor is modelled internally, according to Figure 3.1 and Equations 2.

It is assumed that when the SSSC bypass switch is closed, The PWM converter is blocked and at the same time a separate breaker at the PWM terminals is closed, see LVB breaker in Figure 1.2. In this conditions, PWM diodes cannot conduct.

The *Two-level converter* model can be controlled with the following set of input for balanced and unbalanced RMS and for EMT simulations:

- *Pmr*, *Pmi*
- *Pmd*, *Pmq*, *cosref*, *sinref*

If the input signals *cosref*, *sinref* are not connected, they are internally calculated based on the positive sequence voltage at bus 1.

Additional signals are available:

- *iDC*, DC-current
- *uDC*, DC-voltage
- *iAC_r*, *iAC_i*, AC-current, real and imaginary part
- *ubus1_r*, *ubus1_i*, Bus 1 voltage
- *upwm_r*, *upwm_i*, PWM terminal voltage
- *id*, *iq*, AC-current, d- and q-axis
- *Ucap*, DC capacitor voltage

6 Short-Circuit Complete

The SSSC can be modelled in the complete short-circuit calculation either as bypassed or as a reactance. The reactance can either be the same as in the load flow, or defined separately.

When the option ”Consider bypass circuit” is selected, the SSSC will be bypassed if the short-circuit current is higher than the defined limit.

7 Short-Circuit VDE/IEC/ANSI

The SSSC can be modelled in the VDE/IEC/ANSI short-circuit calculation either as bypassed or as a specified reactance.

Warning! This setting is not part of the VDE0102/IEC60909 or ANSI standards.

8 Harmonics

For harmonics calculations the SSSC is modelled as a voltage source in series with an impedance. A frequency dependent characteristic can be defined for the positive and zero sequence resistance and reactance. The injected voltages at each harmonic can be defined through a Harmonic Voltage (*TypHmcvolt*). The positive, negative and zero sequence model of the SSSC is shown in Figure 8.1.

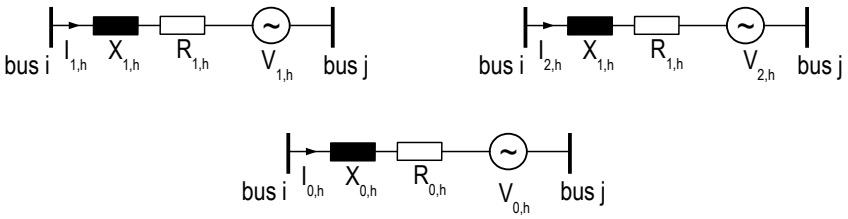


Figure 8.1: Positive, negative and zero sequence harmonics model of the SSSC

The negative sequence impedance is assumed equal to the positive sequence one.

9 Input/Output signals of the RMS and EMT Voltage source model

Name	Description	Unit	Type	Model
bypass	Bypass		IN	EMT, RMS bal. and unbal.
Xin	Reactance setpoint	Ω	IN	RMS bal.
Vin	Voltage setpoint	kV	IN	RMS bal.
Xin_A	Reactance setpoint, phase A	Ω	IN	RMS unbal.
Xin_B	Reactance setpoint, phase B	Ω	IN	RMS unbal.
Xin_C	Reactance setpoint, phase C	Ω	IN	RMS unbal.
Vin_A	Voltage setpoint, phase A	kV	IN	EMT, RMS unbal.
Vin_B	Voltage setpoint, phase B	kV	IN	EMT, RMS unbal.
Vin_C	Voltage setpoint, phase C	kV	IN	EMT, RMS unbal.
u1r_in	Positive sequence voltage input, real part	p.u.	IN	RMS bal. and unbal.
u1i_in	Positive sequence voltage input, imaginary part	p.u.	IN	RMS bal. and unbal.
u2r_in	Negative sequence voltage input, real part	p.u.	IN	RMS unbal.
u2i_in	Negative sequence voltage input, imaginary part	p.u.	IN	RMS unbal.
u0r_in	Zero sequence voltage input, real part	p.u.	IN	RMS unbal.
u0i_in	Zero sequence voltage input, imaginary part	p.u.	IN	RMS unbal.
X	Effective reactance	Ω	OUT	EMT, RMS bal.
X_A	Effective reactance, phase A	Ω	OUT	RMS unbal.
X_B	Effective reactance, phase B	Ω	OUT	RMS unbal.
X_C	Effective reactance, phase C	Ω	OUT	RMS unbal.
V	Effective SSSC voltage	kV	OUT	EMT, RMS bal.
V_A	Effective SSSC voltage, phase A	kV	OUT	EMT, RMS unbal.
V_B	Effective SSSC voltage, phase B	kV	OUT	EMT, RMS unbal.
V_C	Effective SSSC voltage, phase C	kV	OUT	EMT, RMS unbal.

Table 9.1: Input-Output signals of the RMS and EMT Voltage source model

10 Input/Output signals of the RMS and EMT Two-level converter model

Name	Description	Unit	Type	Model
bypass	Bypass		IN	EMT, RMS bal. and unbal.
Pmr	Pulse-modulation factor (real part)		IN	EMT, RMS bal. and unbal.
Pmi	Pulse-modulation factor (imaginary part)		IN	EMT, RMS bal. and unbal.
Pmd	Pulse-modulation factor (d-axis)		IN	EMT, RMS bal. and unbal.
Pmq	Pulse-modulation factor (q-axis)		IN	EMT, RMS bal. and unbal.
cosref	Cos(dq-reference-angle)		IN	EMT, RMS bal. and unbal.
sinref	Sin(dq-reference-angle)		IN	EMT, RMS bal. and unbal.
iDC	DC capacitor current	p.u.	OUT	EMT, RMS bal. and unbal.
uDC	DC capacitor voltage	p.u.	OUT	EMT, RMS bal. and unbal.
iAC_r	AC-current, real part	p.u.	OUT	EMT, RMS bal. and unbal.
iAC_i	AC-current, imaginary part	p.u.	OUT	EMT, RMS bal. and unbal.
ubus1_r	Bus 1 voltage, real part	p.u.	OUT	EMT, RMS bal. and unbal.
ubus1_i	Bus 1 voltage, imaginary part	p.u.	OUT	EMT, RMS bal. and unbal.
upwm_r	PWM terminal voltage, real part	p.u.	OUT	EMT, RMS bal. and unbal.
upwm_i	PWM terminal voltage, imaginary part	p.u.	OUT	EMT, RMS bal. and unbal.
id	AC-current, d-axis	p.u.	OUT	EMT, RMS bal. and unbal.
iq	AC-current, q-axis	p.u.	OUT	EMT, RMS bal. and unbal.

Table 10.1: Input-Output signals of the RMS and EMT Two-level converter model

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